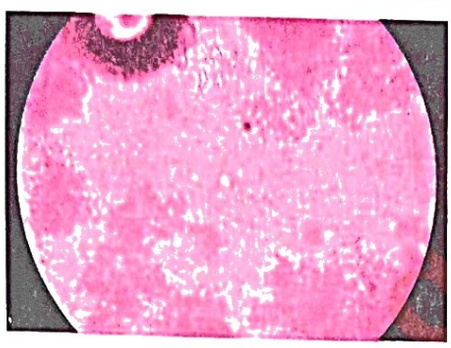
Experiment No. 5

Title : Making the yoghurt from milk in laboratory & study of the bacteria in it under the compound microscope.

Materials : A big conical flask or suitable vessel, glass slide, glass rod, compound microscope.

Chemicals : Milk, yoghurt or butter milk, methylene blue stain, etc.

Figure :



Bacteria as seen under microscope

Procedure :

1. Take about 100 ml of lukewarm milk in a conical flask or a vessel.
2. Add a half or full spoonful of yoghurt or butter milk to it.
3. Put a lid over the conical flask / vessel and keep it in warm place for 8 – 10 hours.
4. Remove the lid from vessel after 8 – 10 hours and observe the material in it.
5. Check the taste, odour & consistency of the yoghurt formed.
6. Take a drop of yoghurt on a glass slide & smear it with the help of other slide.
7. Let the smear dry for some time.
8. Stain the smear with a drop of methylene blue & put a cover slip over it.
9. Observe under the 45X objective of properly adjusted compound microscope.

Observations :

1. Changes during conversion of liquid milk into yoghurt It's taste and oder.
2. Specific taste & odour of yoghurt Sour and Sourer.
3. Microbes seen under the microscope lactobacilli.

Inference :

We can see that from milk we can make yoghurt batter milk or other milk products due to bacteria as seen under microscope is lactobacilli which is a useful microscope used for making milk products.

Multiple Choice Questions

1. Due to ....., yoghurt acquires sour taste.
   A. proteins B. lactic acid C. lactose D. acetic acid
2. During formation of yoghurt from milk, ...
   A. pH decreases B. proteins coagulate
   C. bacteria are destroyed D. all of the above.
3. Which stain is used for observation of lactobacilli?
   A. Eosin. B. Safranin. C. Methylene blue. D. Iodine.
4. Which of the following is wrong statement about lactobacilli?
   A. They are anaerobic.
   B. They fix the nitrogen.
   C. They are used for formation of fermented fodder for cattle.
   D. They are used in production of cocoa.
5. During conversion of milk into yoghurt, ...sugar is converted into acid.
   A. maltose B. lactose C. sucrose D. fructose

Questions

1. How does the process of conversion of milk into yoghurt occur?

The lactabacilli convery lactose these sugar in the milk into lactic an acid also result the OH of milk decreases milk proteins. Thus milk proteins are sp seprated from constuent of milk.

2. Why is the yoghurt very important food material?

milk is one of the very important nutrition it contain protens which is useful for us yoghurt is made up of milk therfor, yoghurt is also useful for our health. Hence yoghurt is very important food and matarial.

3. What is name of process of conversion of milk into yoghurt? Why?

formentation is name of processes of conversion of milk into yoghurt it convert lactose the sugar in milk into lactic acid.

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